T13

1) Длина:

Етимет : $n = 139$ $\sigma_{дл} = 5,722$ мм

Европа : $m = 1000$ $\sigma_{дл} = 6,161$ мм

$$\xi \sim N(\theta_1, \theta_2^2)$$

H_0 : θ_2^2 евр $=$ θ_2^2 етим

H_1 : θ_2^2 евр $\neq$ θ_2^2 етим

$$S^2 = \frac{n}{n-1}\sigma^2$$

S^2 етим $= 32,9785$

S^2 евр $= 37,9959$

$$\tilde{\Delta}_{дл} = \frac{S^2 \text{ етим}}{S^2 \text{ евр}} = 0,86795$$

2) Ширина

Етимет : $n = 139$ $\sigma_ш = 4,612$ мм $\alpha = 0,05$

Европа : $m = 1000$ $\sigma_ш = 3,055$ мм

$$\eta \sim N(\theta_1', \theta_2'^2)$$

H_0 : $\theta_2'^2$ евр $=$ $\theta_2'^2$ етим

$H_1: \quad D_2^{21}\,\text{евр} \neq D_2^{12}\,\text{сим}$

$$S^2 = \frac{n}{n-1}\sigma^2$$

$S^2\,\text{сим} = 21,4246$

$S^2\,\text{евр} = 25,5786$

$$\tilde{\Delta}_m = \frac{S^2\,\text{сим}}{S^2\,\text{евр}} = 0,8376$$

$\tilde{\Delta} \sim F(n-1, m-1) = F(138, 999) \Rightarrow$

$\Rightarrow \quad G: \tilde{\Delta} \leq F_{\frac{\alpha}{2}}(138, 999) \cup \tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}(138, 999)$

$F_{\alpha/2}(138, 999) = 0,76738$

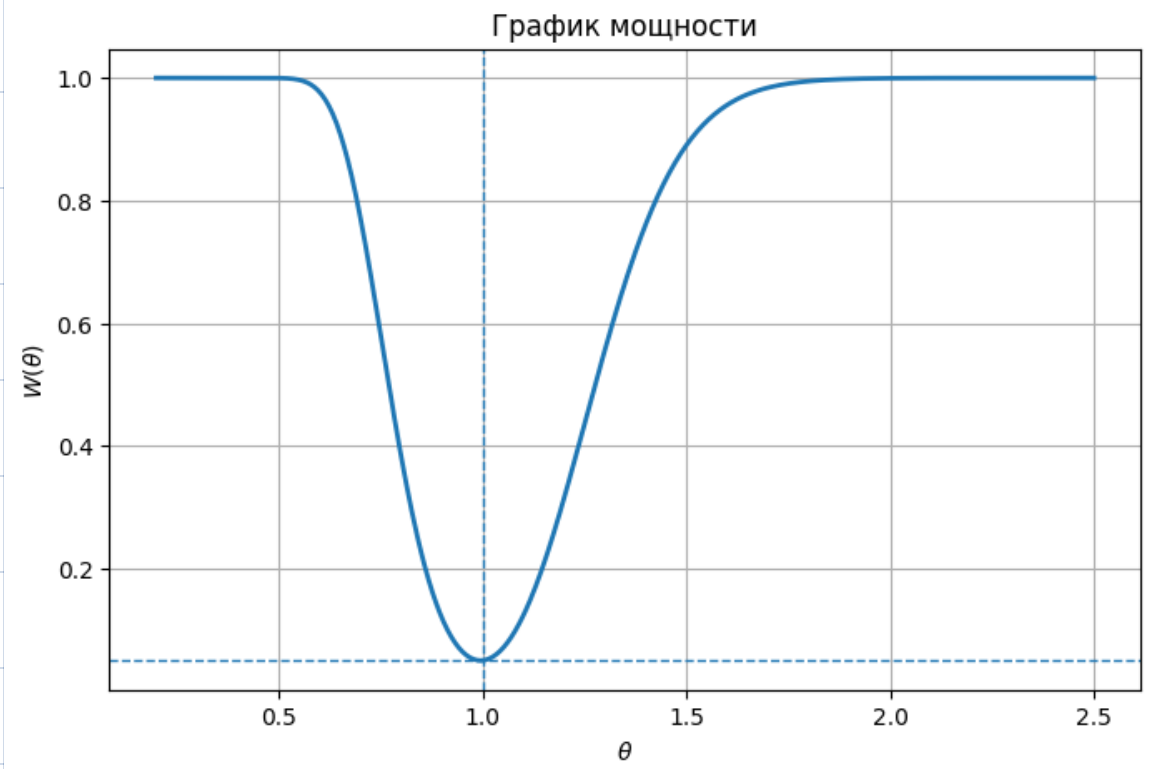
$F_{1-\frac{\alpha}{2}}(138, 999) = 1,27169$

$\Rightarrow \tilde{\Delta}_m$ и $\tilde{\Delta}_{\text{гл}} \notin G \Rightarrow$ Нет оснований отвергать H_0 для 2ух случаев

$$W = P(\Delta \in G \mid H_1) = P\left(\hat{\Delta} \le F_{\frac{\alpha}{2}}(138, 999) \mid H_1\right) +$$

$$+ P\left(\hat{\Delta} \geqslant F_{1-\frac{\alpha}{2}}(138, 999) \mid H_1\right) =$$

$$= P\left(\frac{S^2_{emn}}{S^2_{ebp}} \cdot \frac{\sigma^2_{ebp}}{\sigma^2_{emn}} \le F_{\alpha/2}(138, 999)\right) +$$

$$+ P\left(\frac{S^2_{emn}}{S^2_{ebp}} \cdot \frac{\sigma^2_{ebp}}{\sigma^2_{emn}} \geqslant F_{1-\frac{\alpha}{2}}(138, 999)\right) \Rightarrow$$

$$\Rightarrow W = \int_{F_{\frac{\alpha}{2}}(138, 999)/\theta}^{\infty} q(t)\,dt + \int_{0}^{F_{1-\frac{\alpha}{2}}(138, 999)/\theta} q(t)\,dt$$